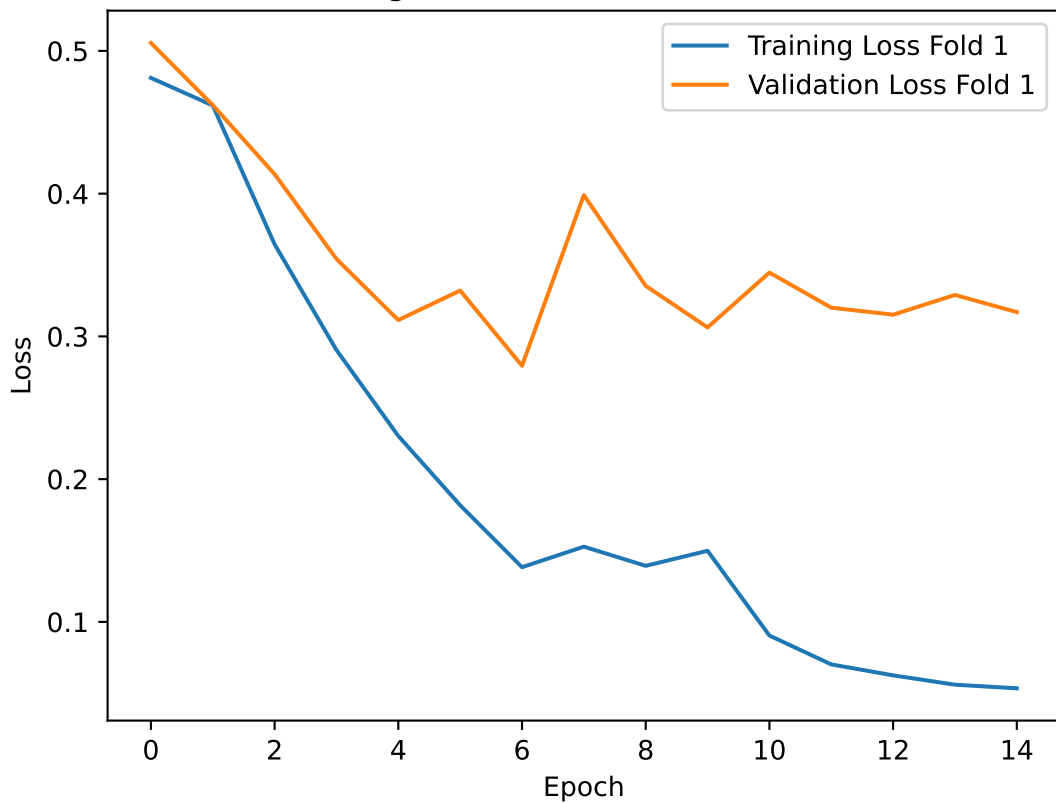
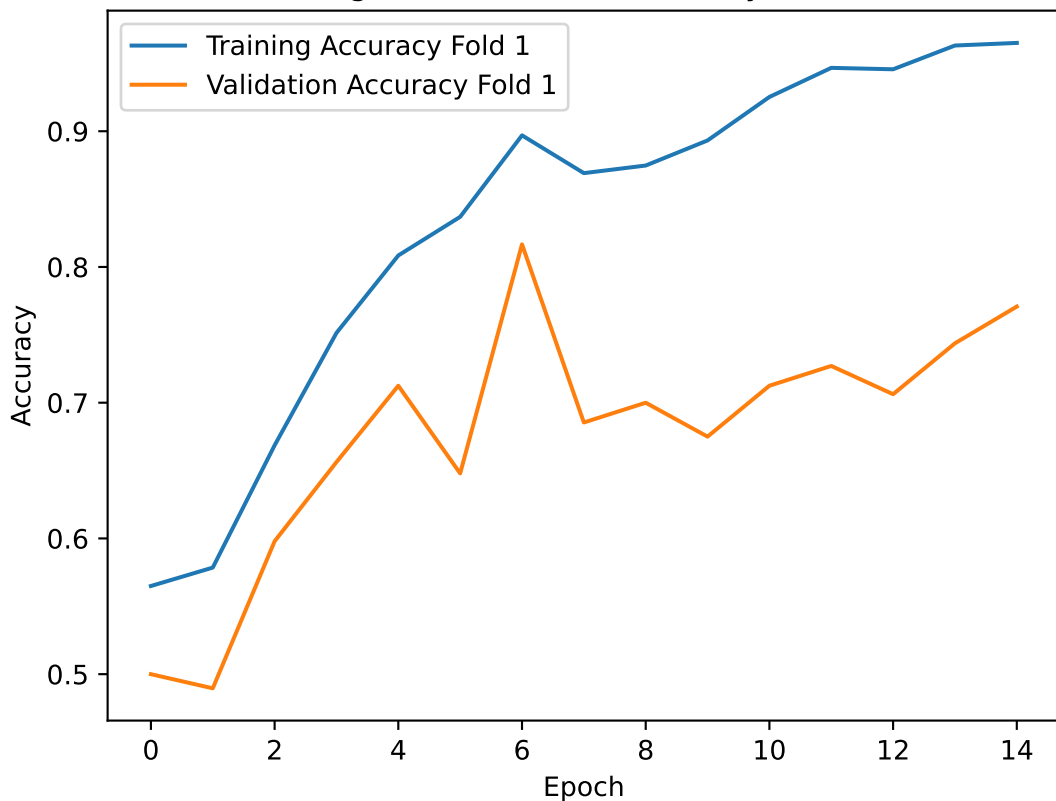


Training and Validation Loss for Fold 1



Training and Validation Accuracy for Fold 1



Best Hyperparameters for Fold 1:

```
{'conv_filters': 64, 'conv_activation': 'relu', 'unitsD1': 288, 'unitsD2': 32}
```

Best Trial Score for Fold 1: 0.7916666865348816

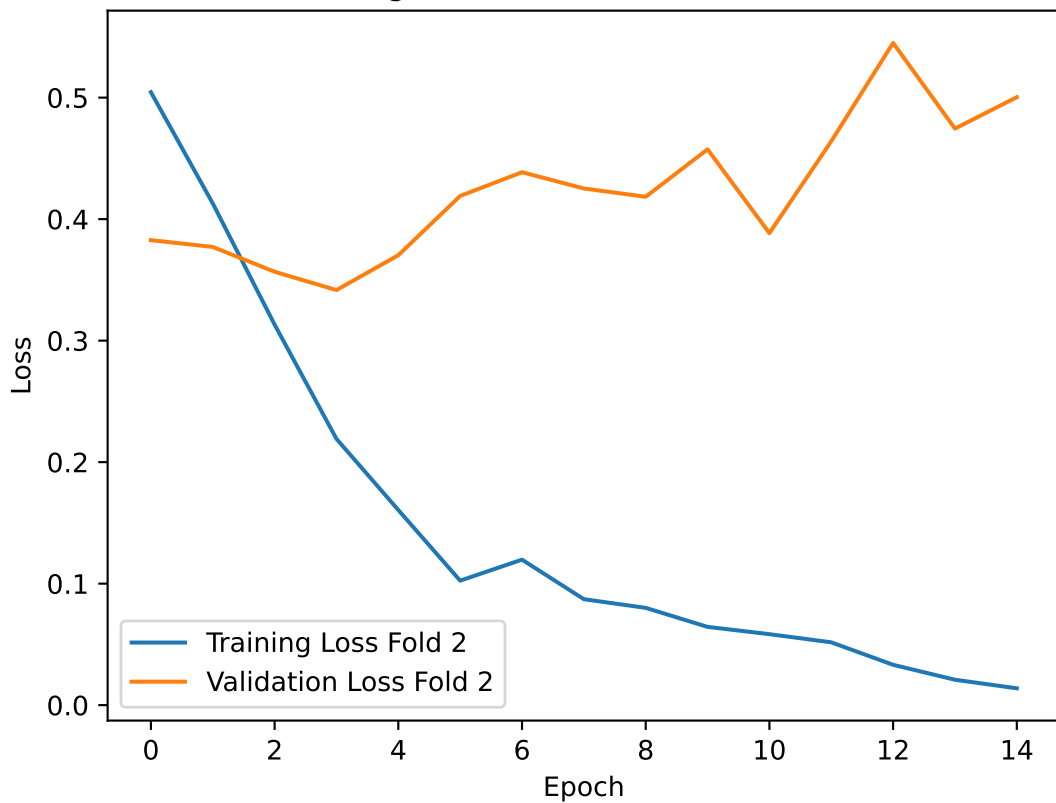
Downloading data from  
https://storage.googleapis.com/tensorflow/keras-applications/resnet/resnet50\_weights\_tf\_dim\_ordering\_tf\_kernels\_notop.h5  
[1m 0/94765736[0m [37m[0m [1m0s[0m  
0s/step[1m 4202496/94765736[0m  
[37m[0m [1m1s[0m  
0us/step[1m15884288/94765736[0m  
[32m[0m[37m[0m [1m0s[0m  
0us/step[1m26484736/94765736[0m  
[32m[0m[37m[0m [1m0s[0m  
0us/step[1m38584320/94765736[0m  
[32m[0m[37m[0m [1m0s[0m  
0us/step[1m44048384/94765736[0m  
[32m[0m[37m[0m [1m0s[0m  
0us/step[1m55803904/94765736[0m  
[32m[0m[37m[0m [1m0s[0m  
0us/step[1m67641344/94765736[0m  
[32m[0m[37m[0m [1m0s[0m  
0us/step[1m80027648/94765736[0m  
[32m[0m[37m[0m [1m0s[0m  
0us/step[1m88088576/94765736[0m  
[32m[0m[37m[0m [1m0s[0m  
0us/step[1m94765736/94765736[0m  
[32m[0m[37m[0m [1m0s[0m 0us/step

Search: Running Trial #1

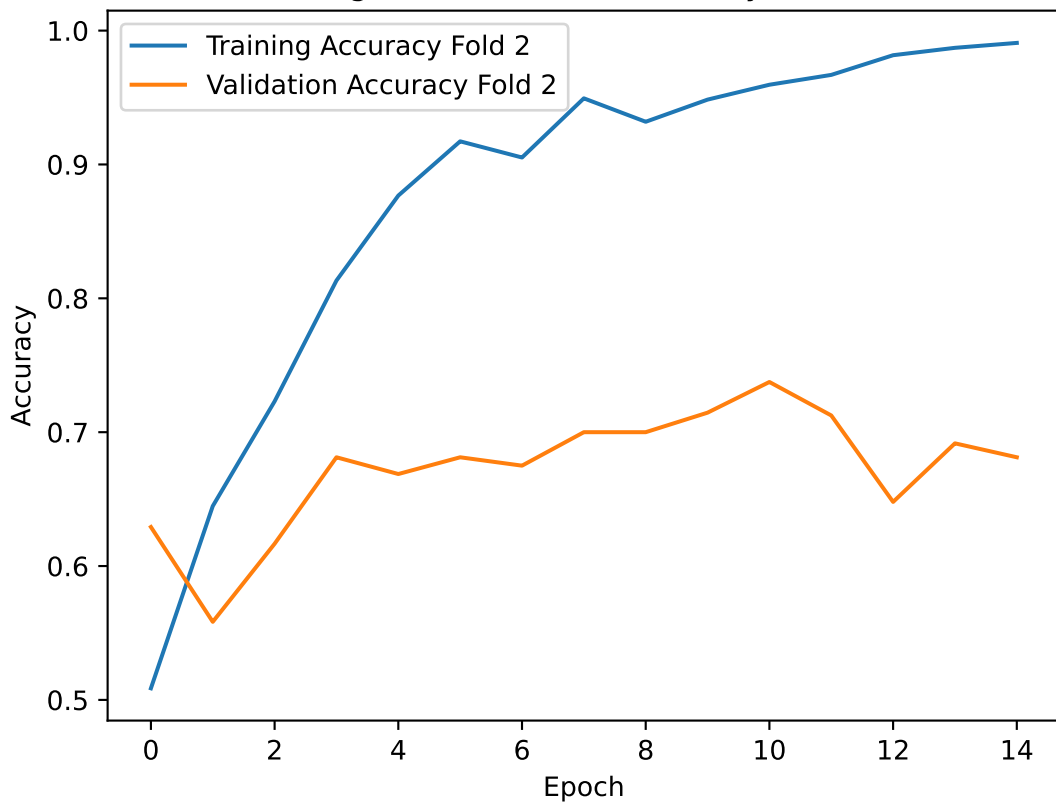
| Value | Best Value So Far | Hyperparameter  |
|-------|-------------------|-----------------|
| 160   | 160               | conv_filters    |
| relu  | relu              | conv_activation |
| 160   | 160               | unitsD1         |
| 32    | 32                | unitsD2         |

Epoch 1/15  
[1m 1/17[0m [32m[0m[37m[0m [1m11:34[0m 43s/step - accuracy: 0.3750 - loss: 0.4661[1m 2/17[0m  
[32m[0m[37m[0m [1m3s[0m 251ms/step - accuracy: 0.4844 - loss: 0.4224  
[1m 3/17[0m  
[32m[0m[37m[0m [1m3s[0m 248ms/step - accuracy: 0.5139 - loss: 0.4184[1m 4/17[0m  
[32m[0m[37m[0m [1m3s[0m 247ms/step - accuracy: 0.5241 - loss: 0.4473[1m 5/17[0m  
[32m[0m[37m[0m [1m2s[0m 249ms/step - accuracy: 0.5268 - loss: 0.4642[1m 6/17[0m  
[32m[0m[37m[0m [1m2s[0m 248ms/step - accuracy: 0.5240 - loss: 0.4770[1m 7/17[0m  
[32m[0m[37m[0m [1m2s[0m 248ms/step - accuracy: 0.5263 - loss: 0.4835[1m 8/17[0m  
[32m[0m[37m[0m [1m2s[0m 248ms/step - accuracy: 0.5245 - loss: 0.4899[1m 9/17[0m  
[32m[0m[37m[0m [1m1s[0m 248ms/step - accuracy: 0.5237 - loss: 0.4938[1m10/17[0m  
[32m[0m[37m[0m [1m1s[0m 248ms/step - accuracy: 0.5214 - loss: 0.4983[1m11/17[0m  
[32m[0m[37m[0m [1m1s[0m 248ms/step - accuracy: 0.5192 - loss: 0.5032[1m12/17[0m  
[32m[0m[37m[0m [1m1s[0m 248ms/step - accuracy: 0.5189 - loss: 0.5059[1m13/17[0m  
[32m[0m[37m[0m [1m0s[0m 248ms/step - accuracy: 0.5194 - loss: 0.5075[1m14/17[0m  
[32m[0m[37m[0m [1m0s[0m 248ms/step - accuracy: 0.5204 - loss: 0.5088[1m15/17[0m  
[32m[0m[37m[0m [1m0s[0m 248ms/step - accuracy: 0.5205 - loss: 0.5112[1m16/17[0m  
[32m[0m[37m[0m [1m0s[0m 248ms/step - accuracy: 0.5205 - loss: 0.5137[1m17/17[0m  
[32m[0m[37m[0m [1m0s[0m 1s/step - accuracy: 0.5199 - loss: 0.5162  
[1m17/17[0m  
[32m[0m[37m[0m [1m81s[0m 2s/step - accuracy: 0.5194 - loss: 0.5184 - val\_accuracy: 0.4479 - val\_loss: 0.4849  
Epoch 2/15  
[1m 1/17[0m [32m[0m[37m[0m [1m4s[0m 303ms/step - accuracy: 0.5000 - loss: 0.5542[1m 2/17[0m  
[32m[0m[37m[0m [1m3s[0m 253ms/step - accuracy: 0.5625 - loss: 0.4784[1m 3/17[0m  
[32m[0m[37m[0m [1m3s[0m 251ms/step - accuracy: 0.5729 - loss: 0.4606[1m 4/17[0m  
[32m[0m[37m[0m [1m3s[0m 250ms/step - accuracy: 0.5684 - loss: 0.4761[1m 5/17[0m  
[32m[0m[37m[0m [1m3s[0m 250ms/step - accuracy: 0.5647 - loss: 0.4850[1m 6/17[0m  
[32m[0m[37m[0m [1m2s[0m 251ms/step - accuracy: 0.5609 - loss: 0.4903[1m 7/17[0m  
[32m[0m[37m[0m [1m2s[0m 251ms/step - accuracy: 0.5611 - loss: 0.4931[1m 8/17[0m

Training and Validation Loss for Fold 2



Training and Validation Accuracy for Fold 2



Best Hyperparameters for Fold 2:

`{'conv_filters': 32, 'conv_activation': 'relu', 'unitsD1': 288, 'unitsD2': 160}`

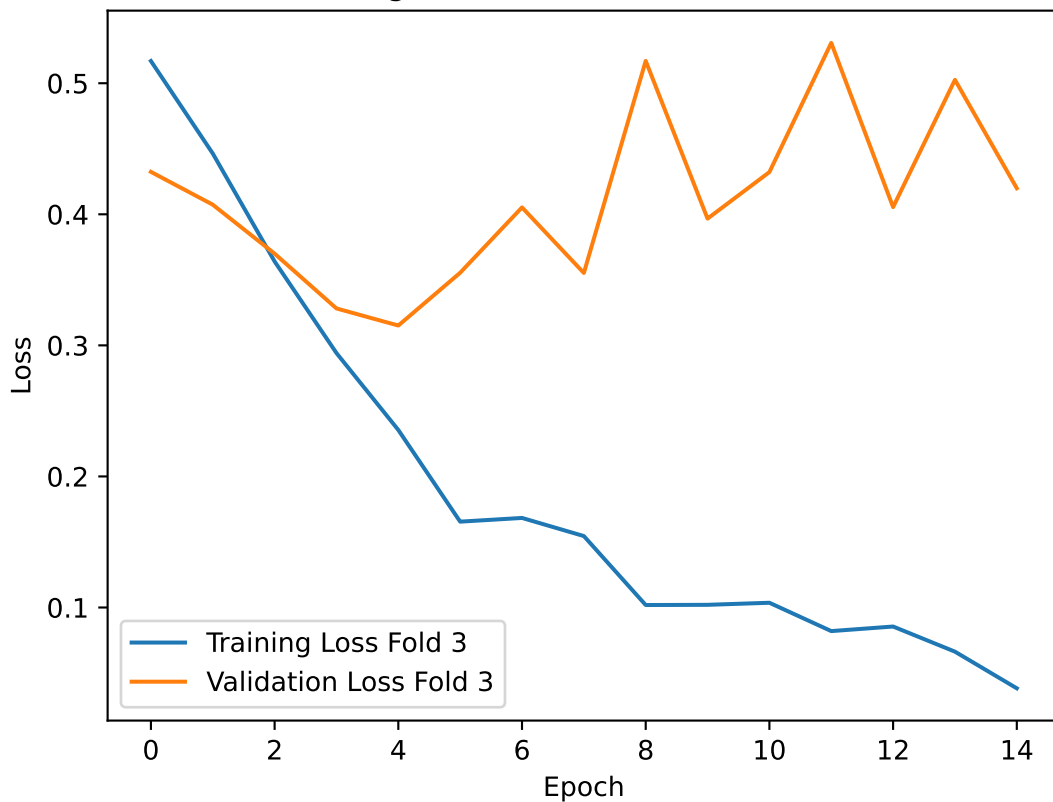
Best Trial Score for Fold 2: 0.7520833611488342

Search: Running Trial #1

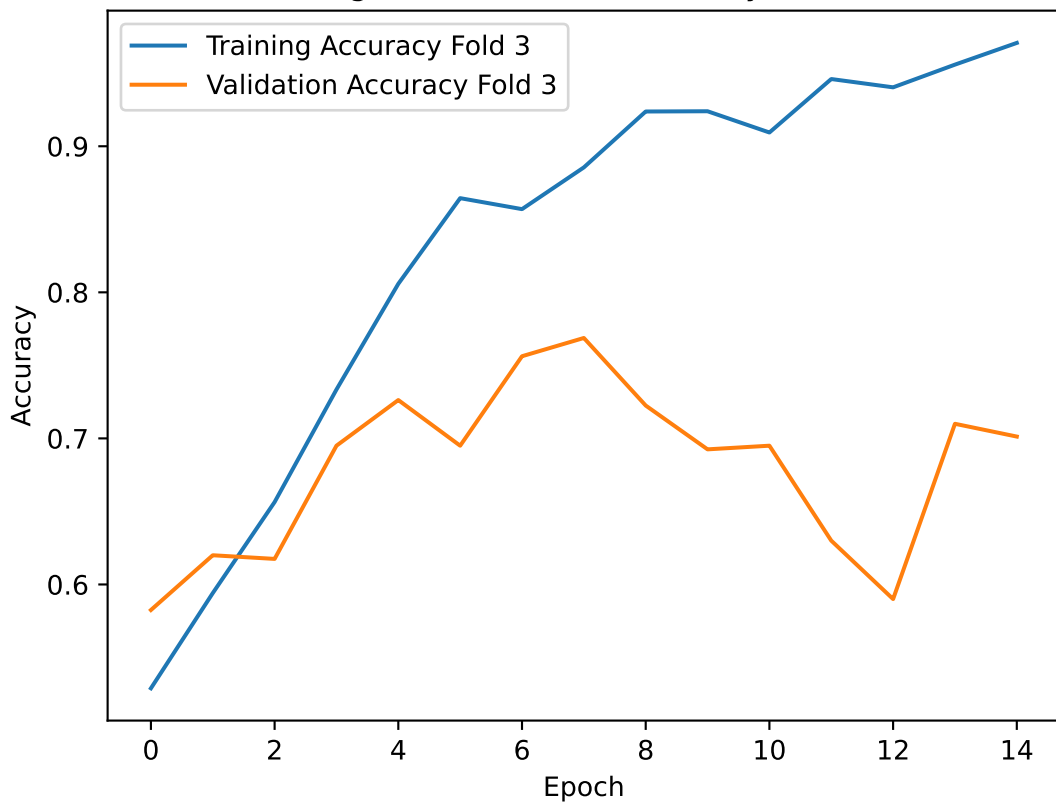
| Value | Best Value So Far | Hyperparameter  |
|-------|-------------------|-----------------|
| 160   | 160               | conv_filters    |
| relu  | relu              | conv_activation |
| 160   | 160               | unitsD1         |
| 32    | 32                | unitsD2         |

Epoch 1/15  
[1m 1/17][0m [32m][0m[37m-----[0m [1m9:55][0m 37s/step - accuracy: 0.4688 - loss: 0.4885  
[32m][0m[37m-----[0m [1m3s][0m 264ms/step - accuracy: 0.5781 - loss: 0.4070  
[32m][0m[37m-----[0m [1m3s][0m 263ms/step - accuracy: 0.6076 - loss: 0.3974  
[32m][0m[37m-----[0m [1m3s][0m 263ms/step - accuracy: 0.5983 - loss: 0.4108  
[32m][0m[37m-----[0m [1m3s][0m 266ms/step - accuracy: 0.5886 - loss: 0.4201  
[32m][0m[37m-----[0m [1m2s][0m 266ms/step - accuracy: 0.5799 - loss: 0.4278  
[32m][0m[37m-----[0m [1m2s][0m 266ms/step - accuracy: 0.5717 - loss: 0.4342  
[32m][0m[37m-----[0m [1m2s][0m 266ms/step - accuracy: 0.5652 - loss: 0.4392  
[32m][0m[37m-----[0m [1m2s][0m 266ms/step - accuracy: 0.5591 - loss: 0.4433  
[32m][0m[37m-----[0m [1m1s][0m 266ms/step - accuracy: 0.5538 - loss: 0.4471  
[32m][0m[37m-----[0m [1m1s][0m 266ms/step - accuracy: 0.5513 - loss: 0.4497  
[32m][0m[37m-----[0m [1m1s][0m 267ms/step - accuracy: 0.5505 - loss: 0.4512  
[32m][0m[37m-----[0m [1m1s][0m 267ms/step - accuracy: 0.5495 - loss: 0.4521  
[32m][0m[37m-----[0m [1m0s][0m 267ms/step - accuracy: 0.5484 - loss: 0.4531  
[32m][0m[37m-----[0m [1m0s][0m 267ms/step - accuracy: 0.5478 - loss: 0.4538  
[32m][0m[37m-----[0m [1m0s][0m 267ms/step - accuracy: 0.5474 - loss: 0.4545  
[32m][0m[37m[0m [1m0s][0m 803ms/step - accuracy: 0.5473 - loss: 0.4552  
[32m][0m[37m[0m [1m69s][0m 2s/step - accuracy: 0.5472 - loss: 0.4558 - val\_accuracy: 0.5375 - val\_loss: 0.4371  
Epoch 2/15  
[1m 1/17][0m [32m][0m[37m-----[0m [1m5s][0m 344ms/step - accuracy: 0.5312 - loss: 0.5141  
[32m][0m[37m-----[0m [1m3s][0m 265ms/step - accuracy: 0.5703 - loss: 0.4850  
[32m][0m[37m-----[0m [1m3s][0m 265ms/step - accuracy: 0.6024 - loss: 0.4557  
[32m][0m[37m-----[0m [1m3s][0m 265ms/step - accuracy: 0.6061 - loss: 0.4528  
[32m][0m[37m-----[0m [1m3s][0m 267ms/step - accuracy: 0.6099 - loss: 0.4513  
[32m][0m[37m-----[0m [1m2s][0m 269ms/step - accuracy: 0.6072 - loss: 0.4553  
[32m][0m[37m-----[0m [1m2s][0m 268ms/step - accuracy: 0.6066 - loss: 0.4588  
[32m][0m[37m-----[0m [1m2s][0m 268ms/step - accuracy: 0.6030 - loss: 0.4646  
[32m][0m[37m-----[0m [1m2s][0m 269ms/step - accuracy: 0.6008 - loss: 0.4682  
[32m][0m[37m-----[0m [1m1s][0m 269ms/step - accuracy: 0.5982 - loss: 0.4709  
[32m][0m[37m-----[0m [1m1s][0m 270ms/step - accuracy: 0.5983 - loss: 0.4703  
[32m][0m[37m-----[0m [1m1s][0m 270ms/step - accuracy: 0.5986 - loss: 0.4694  
[32m][0m[37m-----[0m [1m1s][0m 270ms/step - accuracy: 0.5995 - loss: 0.4682  
[32m][0m[37m-----[0m [1m0s][0m 270ms/step - accuracy: 0.5999 - loss: 0.4675  
[32m][0m[37m-----[0m [1m0s][0m 270ms/step - accuracy: 0.6008 - loss: 0.4661  
[32m][0m[37m-----[0m [1m0s][0m 270ms/step - accuracy: 0.6019 - loss: 0.4647  
[32m][0m[37m[0m [1m0s][0m 265ms/step - accuracy: 0.6029 - loss: 0.4636  
[32m][0m[37m[0m [1m7s][0m 399ms/step - accuracy: 0.6038 - loss: 0.4625 - val\_accuracy: 0.6771 - val\_loss: 0.3793  
Epoch 3/15

Training and Validation Loss for Fold 3



Training and Validation Accuracy for Fold 3



Best Hyperparameters for Fold 3:

```
{'conv_filters': 96, 'conv_activation': 'relu', 'unitsD1': 160, 'unitsD2': 160}
```

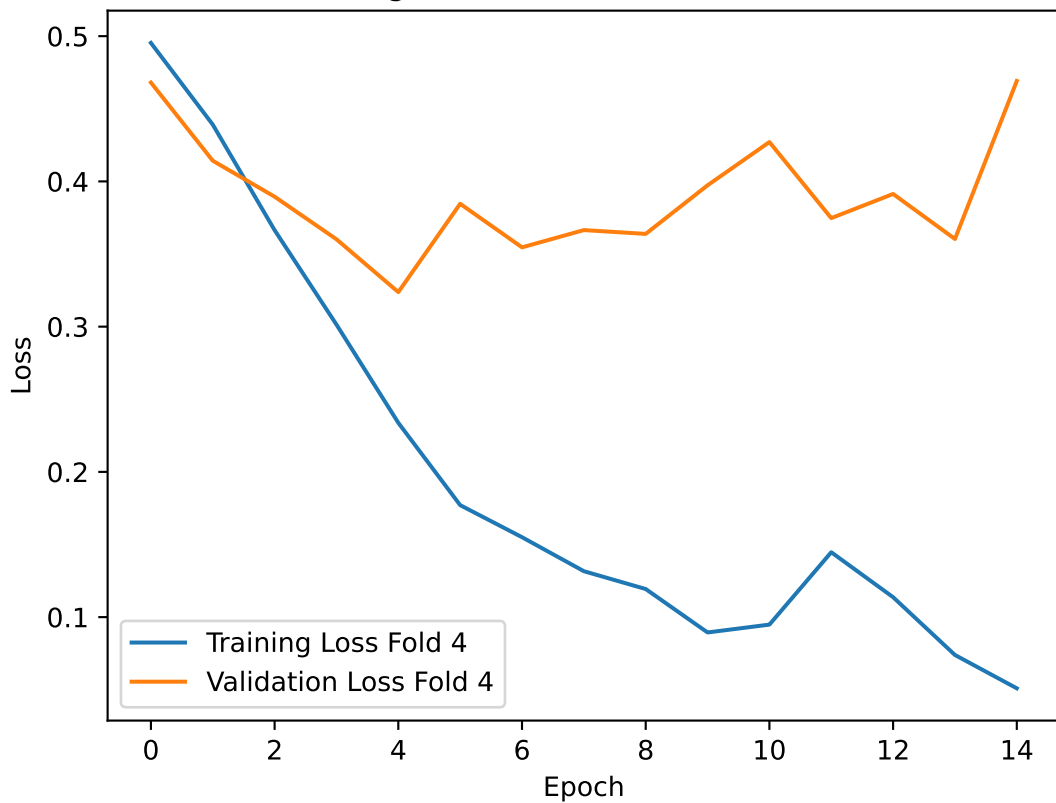
Best Trial Score for Fold 3: 0.7537499666213989

Search: Running Trial #1

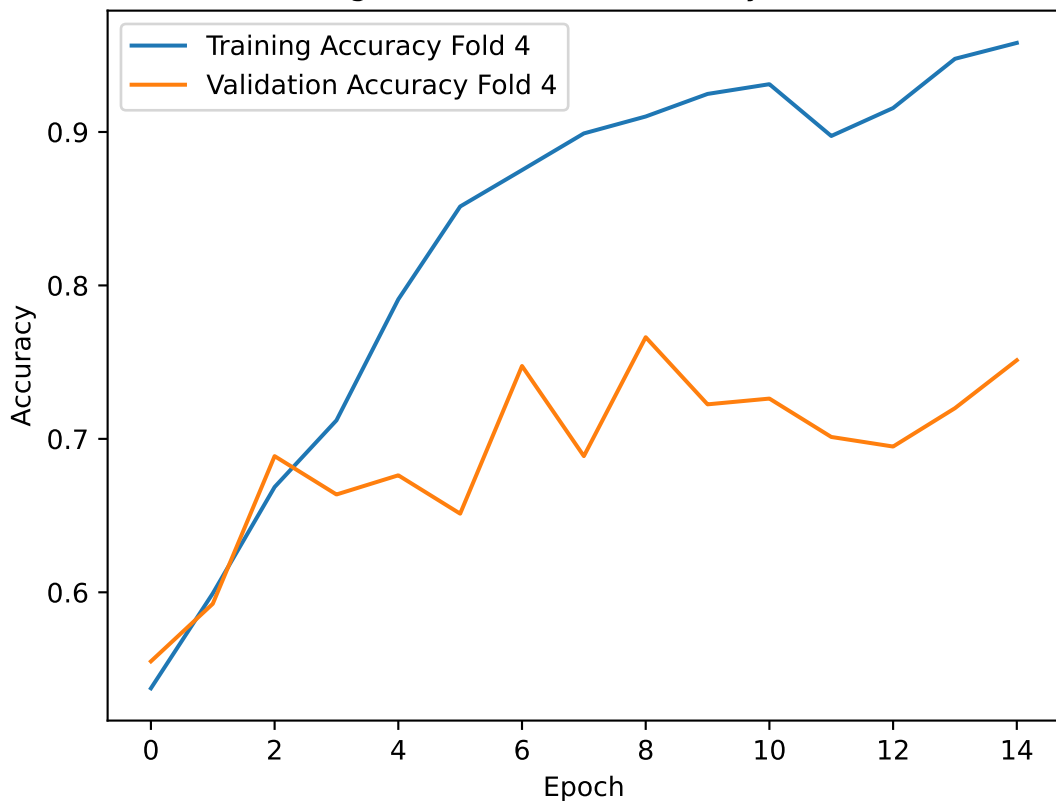
| Value | Best Value So Far | Hyperparameter  |
|-------|-------------------|-----------------|
| 224   | 224               | conv_filters    |
| relu  | relu              | conv_activation |
| 416   | 416               | unitsD1         |
| 160   | 160               | unitsD2         |

Epoch 1/15  
[1m 1/17][0m [32m][0m[37m[0m [1m9:57][0m 37s/step - accuracy: 0.4062 - loss: 0.5602[1m 2/17][0m [32m][0m[37m[0m [1m4s][0m 274ms/step - accuracy: 0.5078 - loss: 0.4795[1m 3/17][0m [32m][0m[37m[0m [1m3s][0m 274ms/step - accuracy: 0.5642 - loss: 0.4345[1m 4/17][0m [32m][0m[37m[0m [1m3s][0m 273ms/step - accuracy: 0.5716 - loss: 0.4341[1m 5/17][0m [32m][0m[37m[0m [1m3s][0m 276ms/step - accuracy: 0.5673 - loss: 0.4356[1m 6/17][0m [32m][0m[37m[0m [1m3s][0m 277ms/step - accuracy: 0.5639 - loss: 0.4392[1m 7/17][0m [32m][0m[37m[0m [1m2s][0m 276ms/step - accuracy: 0.5599 - loss: 0.4427[1m 8/17][0m [32m][0m[37m[0m [1m2s][0m 276ms/step - accuracy: 0.5568 - loss: 0.4453[1m 9/17][0m [32m][0m[37m[0m [1m2s][0m 277ms/step - accuracy: 0.5528 - loss: 0.4483[1m10/17][0m [32m][0m[37m[0m [1m1s][0m 277ms/step - accuracy: 0.5491 - loss: 0.4520[1m11/17][0m [32m][0m[37m[0m [1m1s][0m 277ms/step - accuracy: 0.5449 - loss: 0.4564[1m12/17][0m [32m][0m[37m[0m [1m1s][0m 278ms/step - accuracy: 0.5424 - loss: 0.4594[1m13/17][0m [32m][0m[37m[0m [1m1s][0m 278ms/step - accuracy: 0.5399 - loss: 0.4621[1m14/17][0m [32m][0m[37m[0m [1m0s][0m 278ms/step - accuracy: 0.5382 - loss: 0.4643[1m15/17][0m [32m][0m[37m[0m [1m0s][0m 278ms/step - accuracy: 0.5367 - loss: 0.4659[1m16/17][0m [32m][0m[37m[0m [1m0s][0m 278ms/step - accuracy: 0.5364 - loss: 0.4669[1m17/17][0m [32m][0m[37m[0m [1m0s][0m 1s/step - accuracy: 0.5354 - loss: 0.4681[1m17/17][0m [32m][0m[37m[0m [1m77s][0m 2s/step - accuracy: 0.5344 - loss: 0.4692 - val\_accuracy: 0.5300 - val\_loss: 0.4178  
Epoch 2/15  
[1m 1/17][0m [32m][0m[37m[0m [1m5s][0m 336ms/step - accuracy: 0.4375 - loss: 0.5243[1m 2/17][0m [32m][0m[37m[0m [1m4s][0m 279ms/step - accuracy: 0.5391 - loss: 0.4516[1m 3/17][0m [32m][0m[37m[0m [1m3s][0m 279ms/step - accuracy: 0.5920 - loss: 0.4077[1m 4/17][0m [32m][0m[37m[0m [1m3s][0m 280ms/step - accuracy: 0.6042 - loss: 0.4019[1m 5/17][0m [32m][0m[37m[0m [1m3s][0m 282ms/step - accuracy: 0.6071 - loss: 0.3985[1m 6/17][0m [32m][0m[37m[0m [1m3s][0m 281ms/step - accuracy: 0.6066 - loss: 0.4008[1m 7/17][0m [32m][0m[37m[0m [1m2s][0m 282ms/step - accuracy: 0.6073 - loss: 0.4039[1m 8/17][0m [32m][0m[37m[0m [1m2s][0m 282ms/step - accuracy: 0.6071 - loss: 0.4075[1m 9/17][0m [32m][0m[37m[0m [1m2s][0m 282ms/step - accuracy: 0.6075 - loss: 0.4107[1m10/17][0m [32m][0m[37m[0m [1m1s][0m 282ms/step - accuracy: 0.6065 - loss: 0.4141[1m11/17][0m [32m][0m[37m[0m [1m1s][0m 283ms/step - accuracy: 0.6071 - loss: 0.4160[1m12/17][0m [32m][0m[37m[0m [1m1s][0m 283ms/step - accuracy: 0.6093 - loss: 0.4166[1m13/17][0m [32m][0m[37m[0m [1m1s][0m 283ms/step - accuracy: 0.6105 - loss: 0.4173[1m14/17][0m [32m][0m[37m[0m [1m0s][0m 283ms/step - accuracy: 0.6118 - loss: 0.4172[1m15/17][0m [32m][0m[37m[0m [1m0s][0m 283ms/step - accuracy: 0.6133 - loss: 0.4169[1m16/17][0m [32m][0m[37m[0m [1m0s][0m 283ms/step - accuracy: 0.6151 - loss: 0.4161[1m17/17][0m [32m][0m[37m[0m [1m0s][0m 278ms/step - accuracy: 0.6156 - loss: 0.4161[1m17/17][0m [32m][0m[37m[0m [1m8s][0m 452ms/step - accuracy: 0.6160 - loss: 0.4161 - val\_accuracy: 0.6150 - val\_loss: 0.3601  
Epoch 3/15

Training and Validation Loss for Fold 4



Training and Validation Accuracy for Fold 4



Best Hyperparameters for Fold 4:

```
{'conv_filters': 256, 'conv_activation': 'relu', 'unitsD1': 288, 'unitsD2': 32}
```

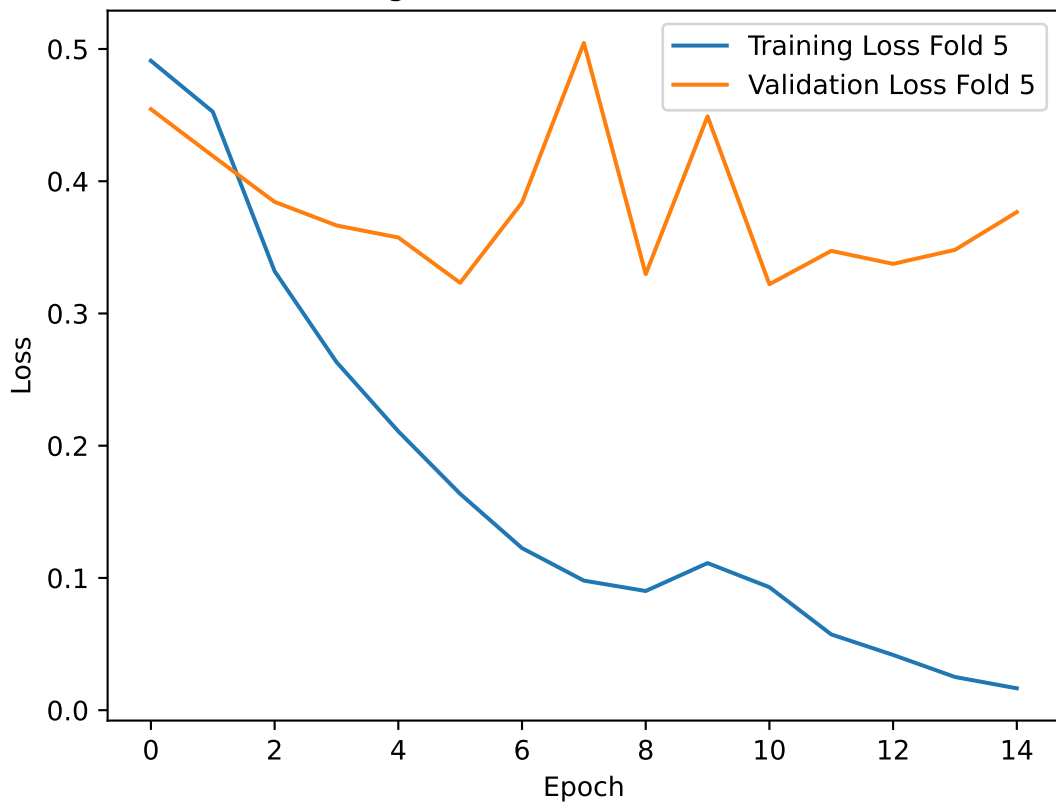
Best Trial Score for Fold 4: 0.8062499761581421

Search: Running Trial #1

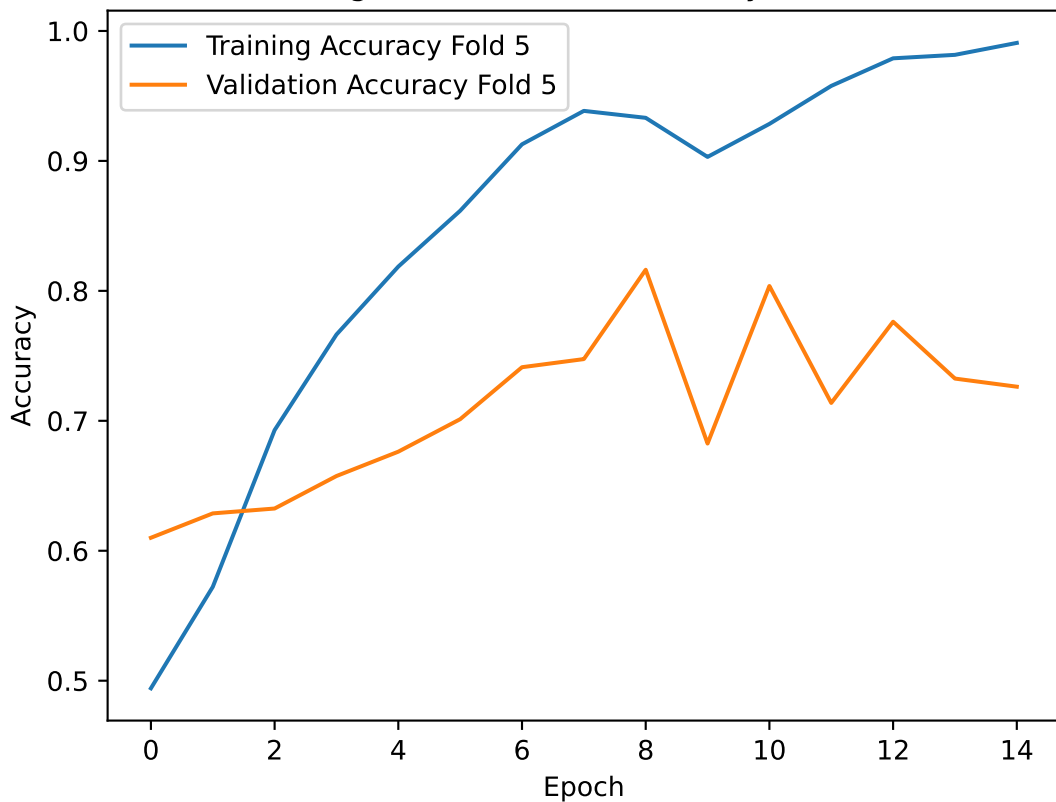
| Value | Best Value So Far | Hyperparameter  |
|-------|-------------------|-----------------|
| 256   | 256               | conv_filters    |
| relu  | relu              | conv_activation |
| 288   | 288               | unitsD1         |
| 32    | 32                | unitsD2         |

Epoch 1/15  
[1m 1/17][0m [32m][0m[37m[0m [1m10:19][0m 39s/step - accuracy: 0.4375 - loss: 0.4914  
[32m][0m[37m[0m [1m4s][0m 273ms/step - accuracy: 0.4766 - loss: 0.4866  
[32m][0m[37m[0m [1m3s][0m 273ms/step - accuracy: 0.5365 - loss: 0.4420  
[32m][0m[37m[0m [1m3s][0m 271ms/step - accuracy: 0.5469 - loss: 0.4716  
[32m][0m[37m[0m [1m3s][0m 274ms/step - accuracy: 0.5412 - loss: 0.4914  
[32m][0m[37m[0m [1m3s][0m 274ms/step - accuracy: 0.5370 - loss: 0.5017  
[32m][0m[37m[0m [1m2s][0m 274ms/step - accuracy: 0.5342 - loss: 0.5075  
[32m][0m[37m[0m [1m2s][0m 273ms/step - accuracy: 0.5339 - loss: 0.5107  
[32m][0m[37m[0m [1m2s][0m 274ms/step - accuracy: 0.5344 - loss: 0.5125  
[32m][0m[37m[0m [1m1s][0m 274ms/step - accuracy: 0.5359 - loss: 0.5129  
[32m][0m[37m[0m [1m1s][0m 274ms/step - accuracy: 0.5368 - loss: 0.5162  
[32m][0m[37m[0m [1m1s][0m 274ms/step - accuracy: 0.5374 - loss: 0.5187  
[32m][0m[37m[0m [1m1s][0m 275ms/step - accuracy: 0.5379 - loss: 0.5204  
[32m][0m[37m[0m [1m0s][0m 275ms/step - accuracy: 0.5379 - loss: 0.5219  
[32m][0m[37m[0m [1m0s][0m 275ms/step - accuracy: 0.5377 - loss: 0.5232  
[32m][0m[37m[0m [1m0s][0m 275ms/step - accuracy: 0.5377 - loss: 0.5247  
[32m][0m[37m[0m [1m0s][0m 922ms/step - accuracy: 0.5370 - loss: 0.5263  
[32m][0m[37m[0m [1m74s][0m 2s/step - accuracy: 0.5365 - loss: 0.5277 - val\_accuracy: 0.5613 - val\_loss: 0.4467  
Epoch 2/15  
[1m 1/17][0m [32m][0m[37m[0m [1m5s][0m 357ms/step - accuracy: 0.5312 - loss: 0.4899  
[32m][0m[37m[0m [1m4s][0m 273ms/step - accuracy: 0.5547 - loss: 0.4684  
[32m][0m[37m[0m [1m3s][0m 274ms/step - accuracy: 0.5955 - loss: 0.4266  
[32m][0m[37m[0m [1m3s][0m 273ms/step - accuracy: 0.5990 - loss: 0.4276  
[32m][0m[37m[0m [1m3s][0m 277ms/step - accuracy: 0.6029 - loss: 0.4300  
[32m][0m[37m[0m [1m3s][0m 277ms/step - accuracy: 0.6040 - loss: 0.4330  
[32m][0m[37m[0m [1m2s][0m 278ms/step - accuracy: 0.6070 - loss: 0.4342  
[32m][0m[37m[0m [1m2s][0m 277ms/step - accuracy: 0.6068 - loss: 0.4352  
[32m][0m[37m[0m [1m2s][0m 277ms/step - accuracy: 0.6050 - loss: 0.4364  
[32m][0m[37m[0m [1m1s][0m 278ms/step - accuracy: 0.6045 - loss: 0.4371  
[32m][0m[37m[0m [1m1s][0m 278ms/step - accuracy: 0.6030 - loss: 0.4383  
[32m][0m[37m[0m [1m1s][0m 278ms/step - accuracy: 0.6026 - loss: 0.4388  
[32m][0m[37m[0m [1m1s][0m 278ms/step - accuracy: 0.6025 - loss: 0.4390  
[32m][0m[37m[0m [1m0s][0m 279ms/step - accuracy: 0.6022 - loss: 0.4395  
[32m][0m[37m[0m [1m0s][0m 279ms/step - accuracy: 0.6019 - loss: 0.4401  
[32m][0m[37m[0m [1m0s][0m 279ms/step - accuracy: 0.6015 - loss: 0.4409  
[32m][0m[37m[0m [1m0s][0m 274ms/step - accuracy: 0.6009 - loss: 0.4419  
[32m][0m[37m[0m [1m7s][0m 426ms/step - accuracy: 0.6003 - loss: 0.4427 - val\_accuracy: 0.6012 - val\_loss: 0.4025  
Epoch 3/15

Training and Validation Loss for Fold 5



Training and Validation Accuracy for Fold 5



Best Hyperparameters for Fold 5:

```
{'conv_filters': 256, 'conv_activation': 'relu', 'unitsD1': 416, 'unitsD2': 32}
```

Best Trial Score for Fold 5: 0.7975000143051147

Search: Running Trial #1

| Value | Best Value So Far | Hyperparameter  |
|-------|-------------------|-----------------|
| 256   | 256               | conv_filters    |
| relu  | relu              | conv_activation |
| 416   | 416               | unitsD1         |
| 32    | 32                | unitsD2         |

Epoch 1/15  
[1m 1/17][0m [32m][0m][37m-----][0m [1m10:22][0m 39s/step - accuracy: 0.2500 - loss: 0.5124][1m 2/17][0m [32m][0m][37m-----][0m [1m4s][0m 280ms/step - accuracy: 0.4062 - loss: 0.4289][1m 3/17][0m [32m][0m][37m-----][0m [1m3s][0m 277ms/step - accuracy: 0.4688 - loss: 0.4153][1m 4/17][0m [32m][0m][37m-----][0m [1m3s][0m 276ms/step - accuracy: 0.4902 - loss: 0.4247][1m 5/17][0m [32m][0m][37m-----][0m [1m3s][0m 278ms/step - accuracy: 0.5009 - loss: 0.4328][1m 6/17][0m [32m][0m][37m-----][0m [1m3s][0m 277ms/step - accuracy: 0.5025 - loss: 0.4415][1m 7/17][0m [32m][0m][37m-----][0m [1m2s][0m 277ms/step - accuracy: 0.5066 - loss: 0.4476][1m 8/17][0m [32m][0m][37m-----][0m [1m2s][0m 277ms/step - accuracy: 0.5097 - loss: 0.4520][1m 9/17][0m [32m][0m][37m-----][0m [1m2s][0m 277ms/step - accuracy: 0.5106 - loss: 0.4564][1m10/17][0m [32m][0m][37m-----][0m [1m1s][0m 277ms/step - accuracy: 0.5101 - loss: 0.4607][1m11/17][0m [32m][0m][37m-----][0m [1m1s][0m 277ms/step - accuracy: 0.5115 - loss: 0.4638][1m12/17][0m [32m][0m][37m-----][0m [1m1s][0m 277ms/step - accuracy: 0.5132 - loss: 0.4659][1m13/17][0m [32m][0m][37m-----][0m [1m1s][0m 277ms/step - accuracy: 0.5151 - loss: 0.4678][1m14/17][0m [32m][0m][37m-----][0m [1m0s][0m 277ms/step - accuracy: 0.5163 - loss: 0.4702][1m15/17][0m [32m][0m][37m-----][0m [1m0s][0m 278ms/step - accuracy: 0.5168 - loss: 0.4730][1m16/17][0m [32m][0m][37m-----][0m [1m0s][0m 277ms/step - accuracy: 0.5173 - loss: 0.4761][1m17/17][0m [32m][0m][37m][0m [1m0s][0m 887ms/step - accuracy: 0.5164 - loss: 0.4790][1m17/17][0m [32m][0m][37m][0m [1m74s][0m 2s/step - accuracy: 0.5157 - loss: 0.4817 - val\_accuracy: 0.5613 - val\_loss: 0.4336  
Epoch 2/15  
[1m 1/17][0m [32m][0m][37m-----][0m [1m5s][0m 349ms/step - accuracy: 0.4375 - loss: 0.4807][1m 2/17][0m [32m][0m][37m-----][0m [1m4s][0m 276ms/step - accuracy: 0.5625 - loss: 0.4064][1m 3/17][0m [32m][0m][37m-----][0m [1m3s][0m 276ms/step - accuracy: 0.6111 - loss: 0.3759][1m 4/17][0m [32m][0m][37m-----][0m [1m3s][0m 275ms/step - accuracy: 0.6243 - loss: 0.3708][1m 5/17][0m [32m][0m][37m-----][0m [1m3s][0m 278ms/step - accuracy: 0.6282 - loss: 0.3717][1m 6/17][0m [32m][0m][37m-----][0m [1m3s][0m 279ms/step - accuracy: 0.6251 - loss: 0.3790][1m 7/17][0m [32m][0m][37m-----][0m [1m2s][0m 279ms/step - accuracy: 0.6244 - loss: 0.3853][1m 8/17][0m [32m][0m][37m-----][0m [1m2s][0m 278ms/step - accuracy: 0.6226 - loss: 0.3917][1m 9/17][0m [32m][0m][37m-----][0m [1m2s][0m 279ms/step - accuracy: 0.6221 - loss: 0.3965][1m10/17][0m [32m][0m][37m-----][0m [1m1s][0m 279ms/step - accuracy: 0.6211 - loss: 0.4009][1m11/17][0m [32m][0m][37m-----][0m [1m1s][0m 279ms/step - accuracy: 0.6217 - loss: 0.4038][1m12/17][0m [32m][0m][37m-----][0m [1m1s][0m 279ms/step - accuracy: 0.6233 - loss: 0.4054][1m13/17][0m [32m][0m][37m-----][0m [1m1s][0m 280ms/step - accuracy: 0.6247 - loss: 0.4064][1m14/17][0m [32m][0m][37m-----][0m [1m0s][0m 280ms/step - accuracy: 0.6259 - loss: 0.4072][1m15/17][0m [32m][0m][37m-----][0m [1m0s][0m 280ms/step - accuracy: 0.6268 - loss: 0.4079][1m16/17][0m [32m][0m][37m-----][0m [1m0s][0m 280ms/step - accuracy: 0.6276 - loss: 0.4084][1m17/17][0m [32m][0m][37m][0m [1m0s][0m 275ms/step - accuracy: 0.6276 - loss: 0.4091][1m17/17][0m [32m][0m][37m][0m [1m7s][0m 446ms/step - accuracy: 0.6276 - loss: 0.4097 - val\_accuracy: 0.6725 - val\_loss: 0.4265  
Epoch 3/15

Average Validation F1-Score across all folds:

0.6574712643678161

Average Validation MSE across all folds:

0.3921058050791423

Average Validation Accuracy across all folds:

0.6857666607697805

Best Hyperparameters Found:

Best Fold: 4

Best Trial Score: 0.8062499761581421

Best Hyperparameters: {'conv\_filters': 256, 'conv\_activation': 'relu', 'units': 128, 'dense\_activation': 'relu', 'dense\_units': 128, 'dropout': 0.5, 'optimizer': 'adam', 'loss': 'categorical\_crossentropy', 'metrics': ['accuracy']}